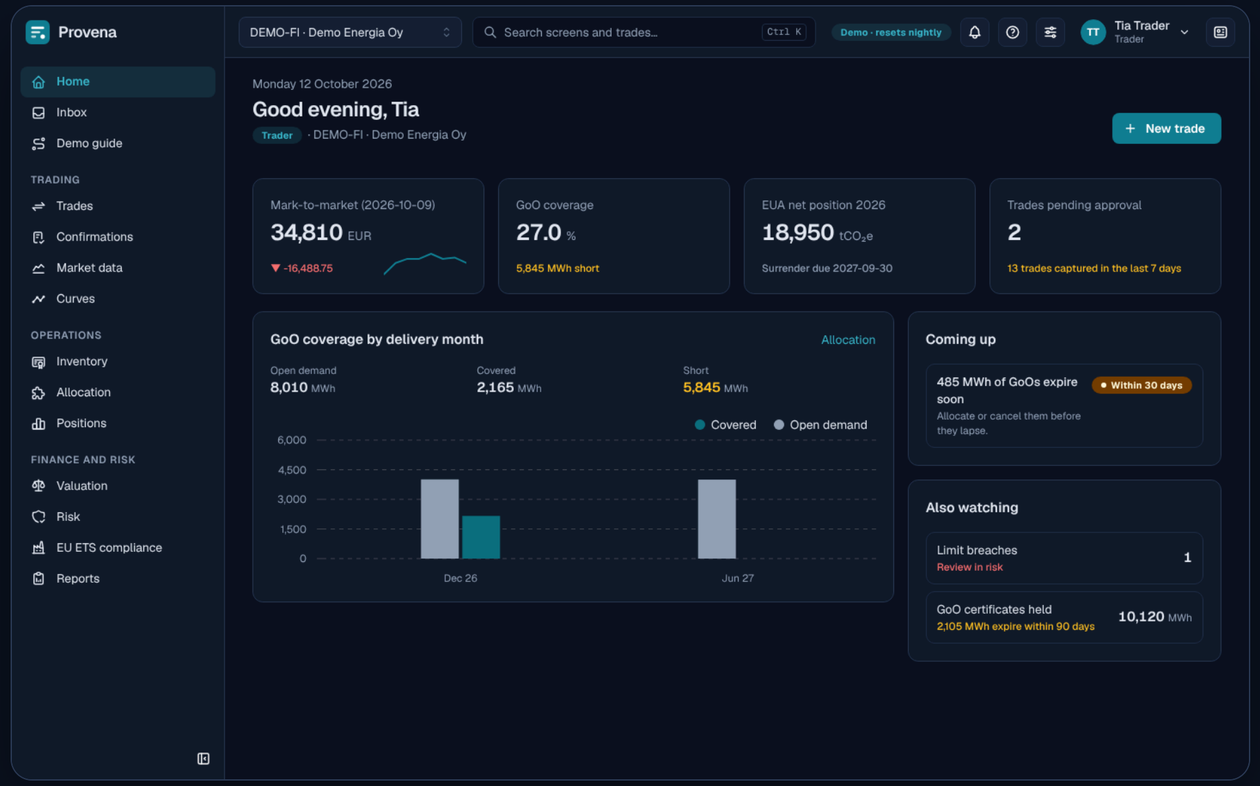
PROVENA

Certificates and allowances, from trade to ledger

A trading and risk system for guarantees of origin and EU emission allowances, built for audited, regulated operations.



THE PROBLEM

Why this is hard

Certificates are ranges

A guarantee of origin is a range of numbers with attributes: country, source, technology, age and support.

Sales name criteria

A sale says which certificates qualify, not which lots; any lot that matches can deliver.

Registries differ

The Union Registry, national GoO registries and the AIB hub, through files, APIs and manual steps.

Two people, fixed dates

Every step that moves value needs a second person, and compliance deadlines are set by law.

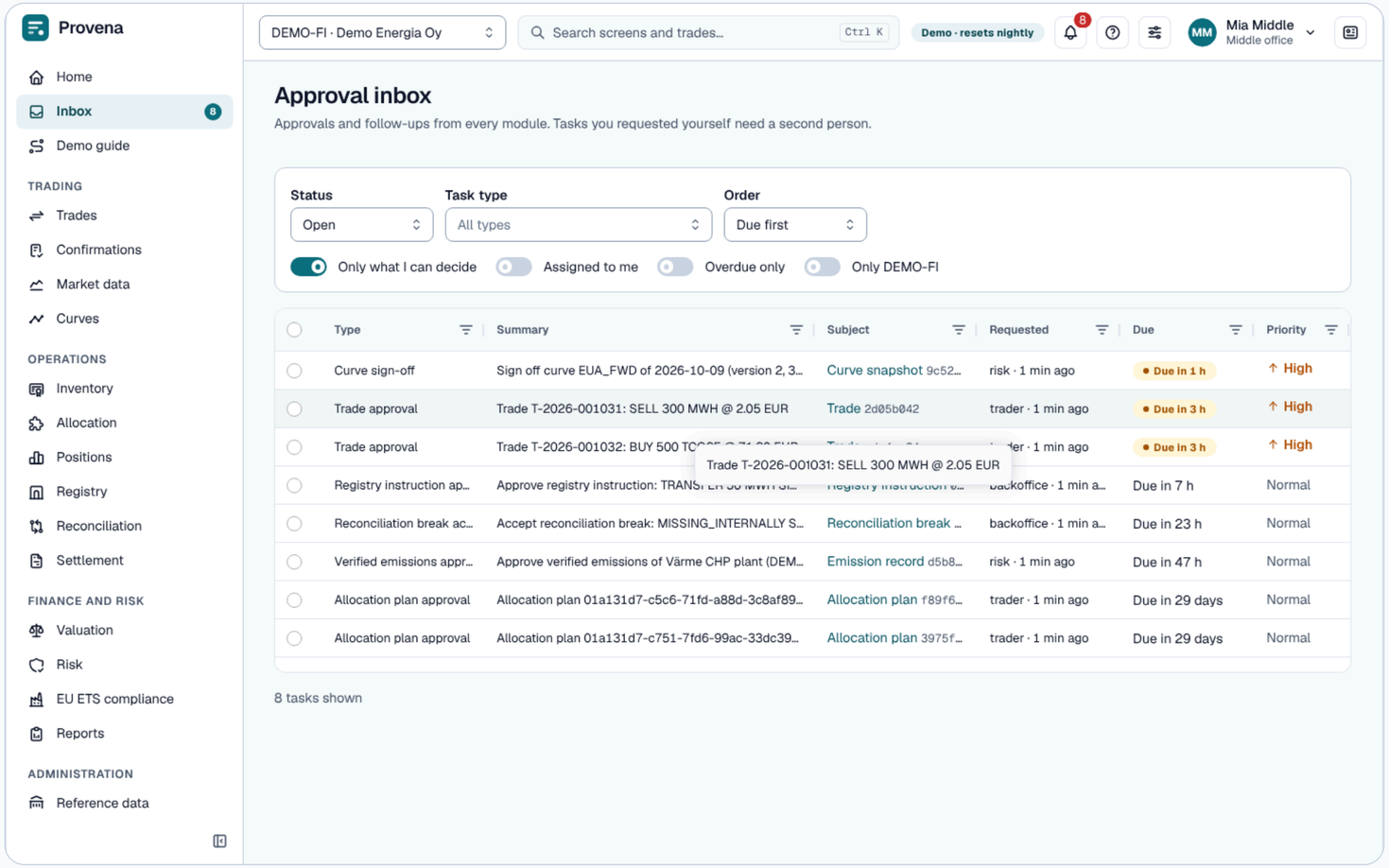
ONE SYSTEM

Every role, one inbox

Trading, operations, finance, risk and administration in one application.

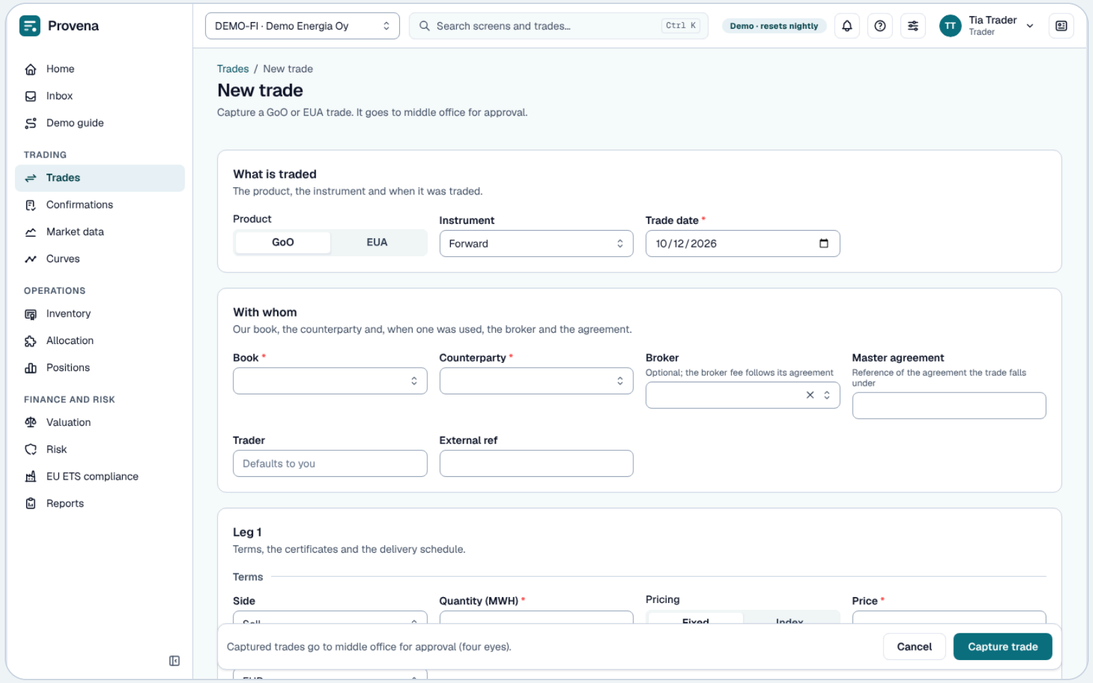
One inbox for every approval, with its four-eyes checks shown.

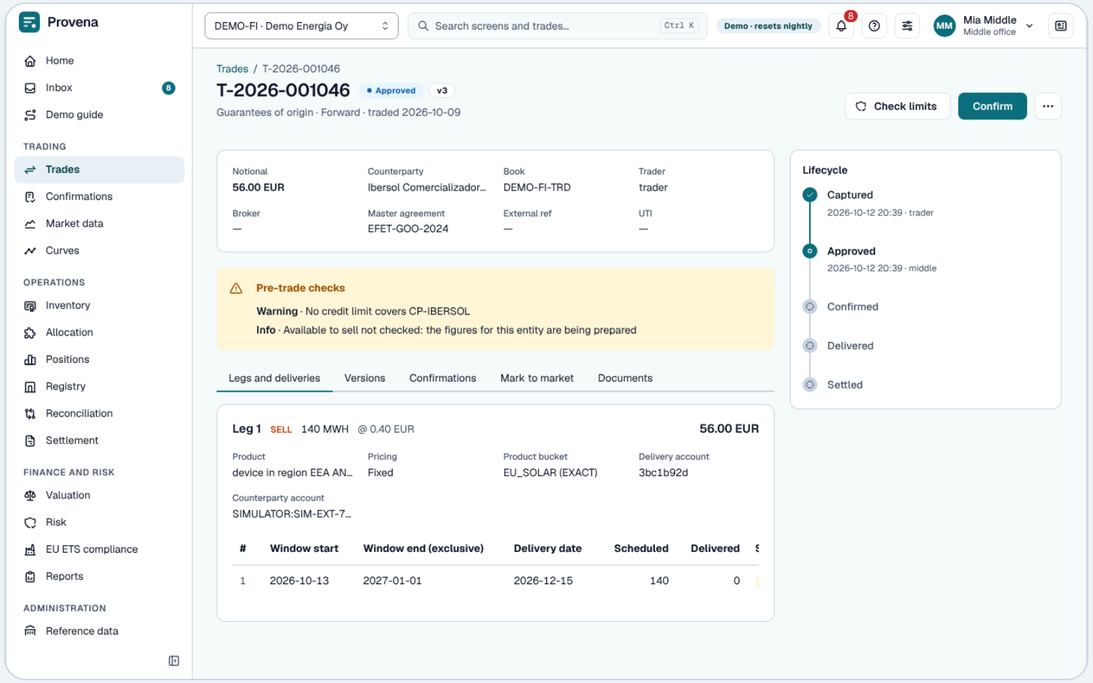
Each role has its own home page and its own work.



TRADING

Capture, approve, confirm, deliver





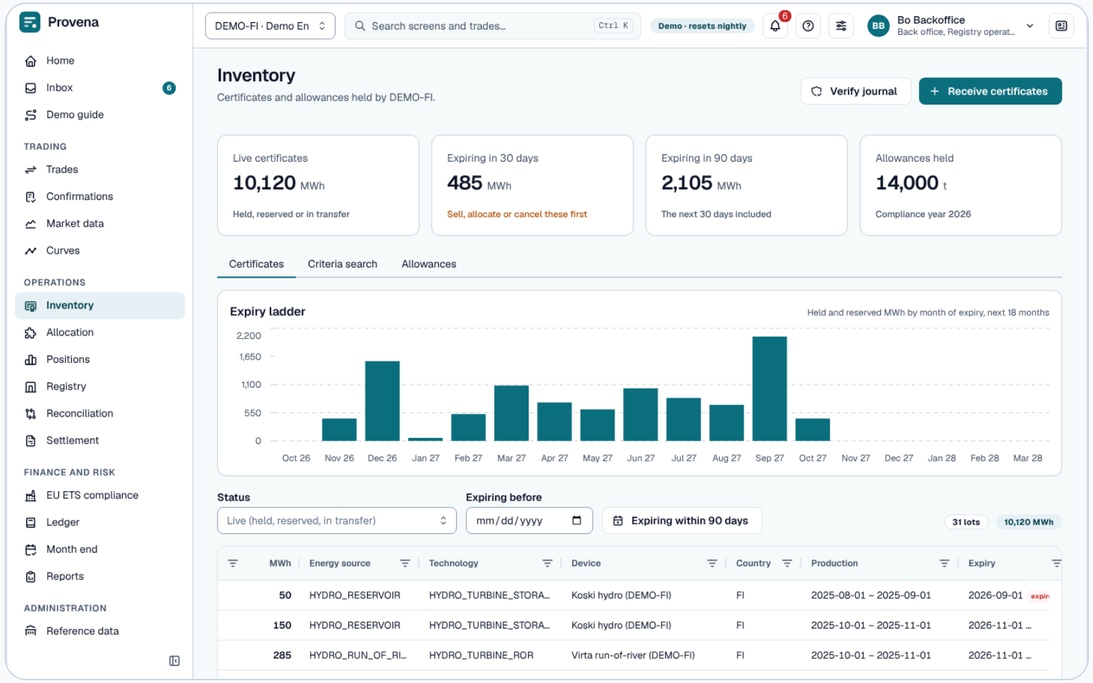
Forwards, spots, futures and options; fixed or index prices.

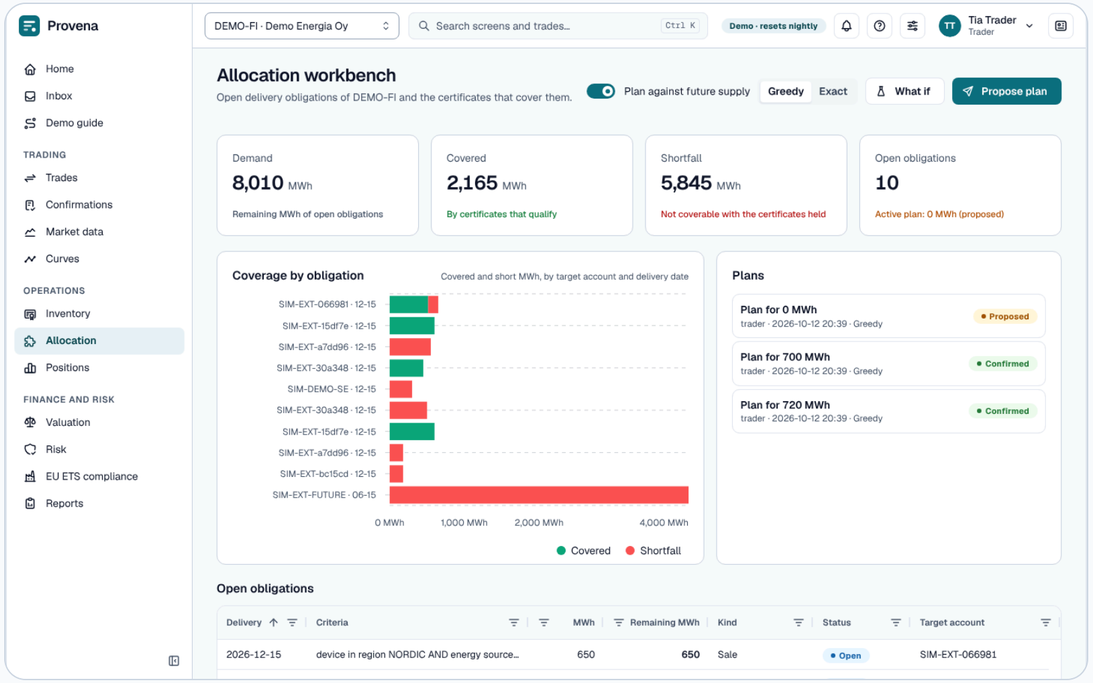
Pre-trade limit checks, then approval by a second person.

Confirmations matched field by field; delivery and settlement.

INVENTORY AND ALLOCATION

The right certificates for every sale





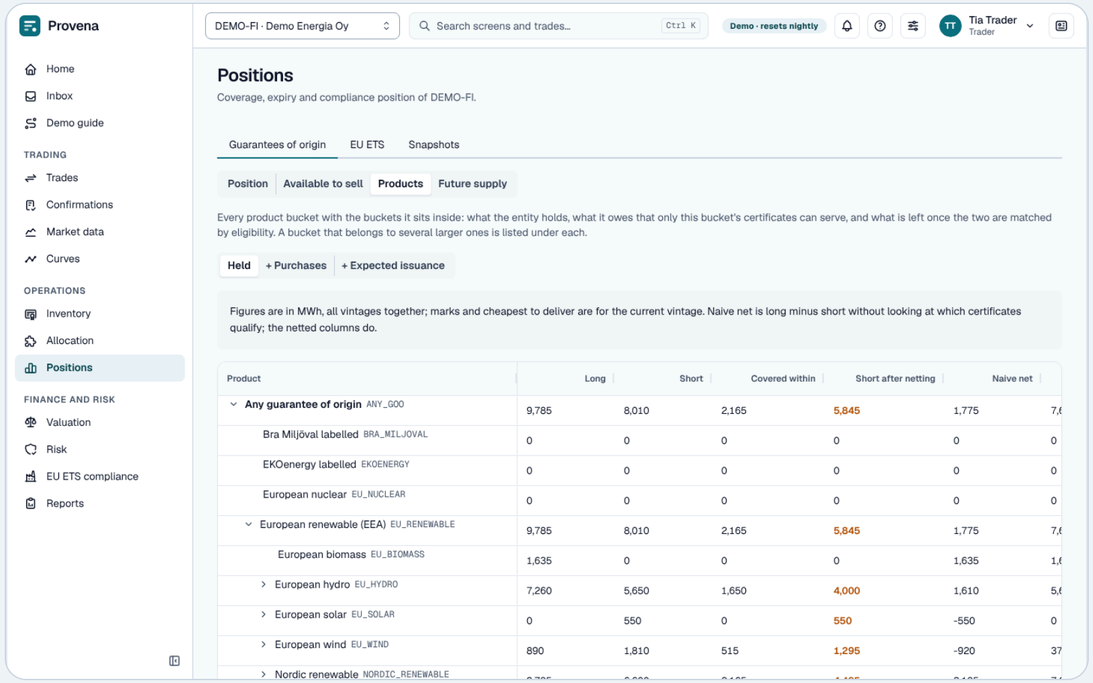
Lots with all their attributes and a hash-chained journal.

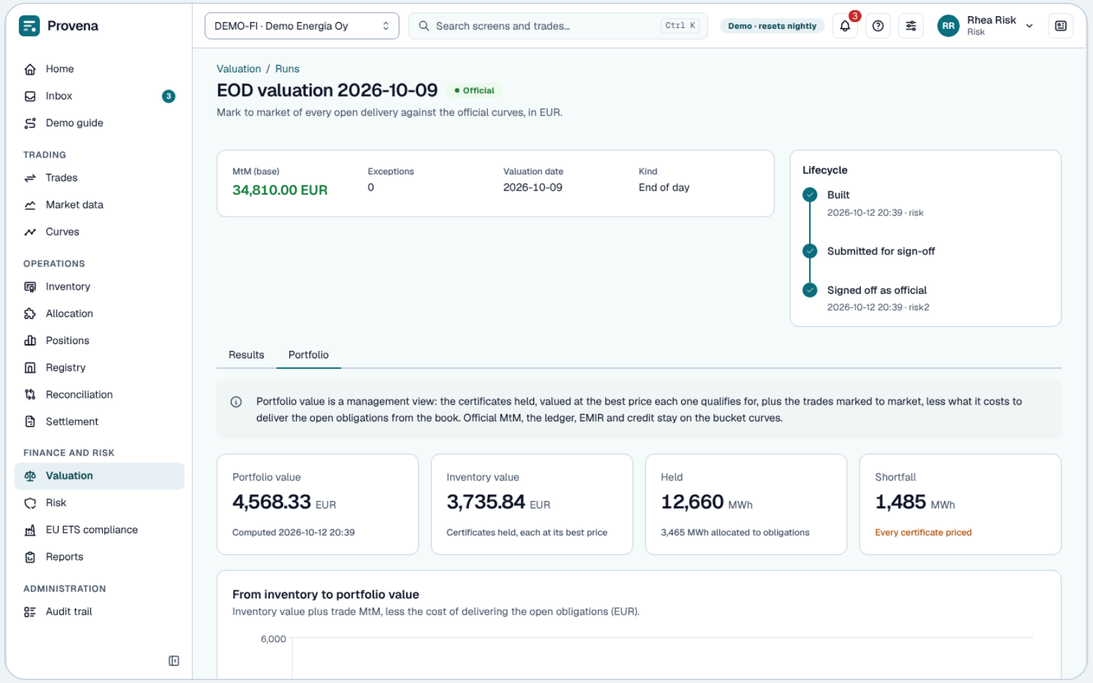
One criteria builder for trades and searches.

Greedy or exact (CP-SAT) allocation, with what-if.

GOO PORTFOLIO MANAGEMENT

The book as a portfolio, not a list of lots





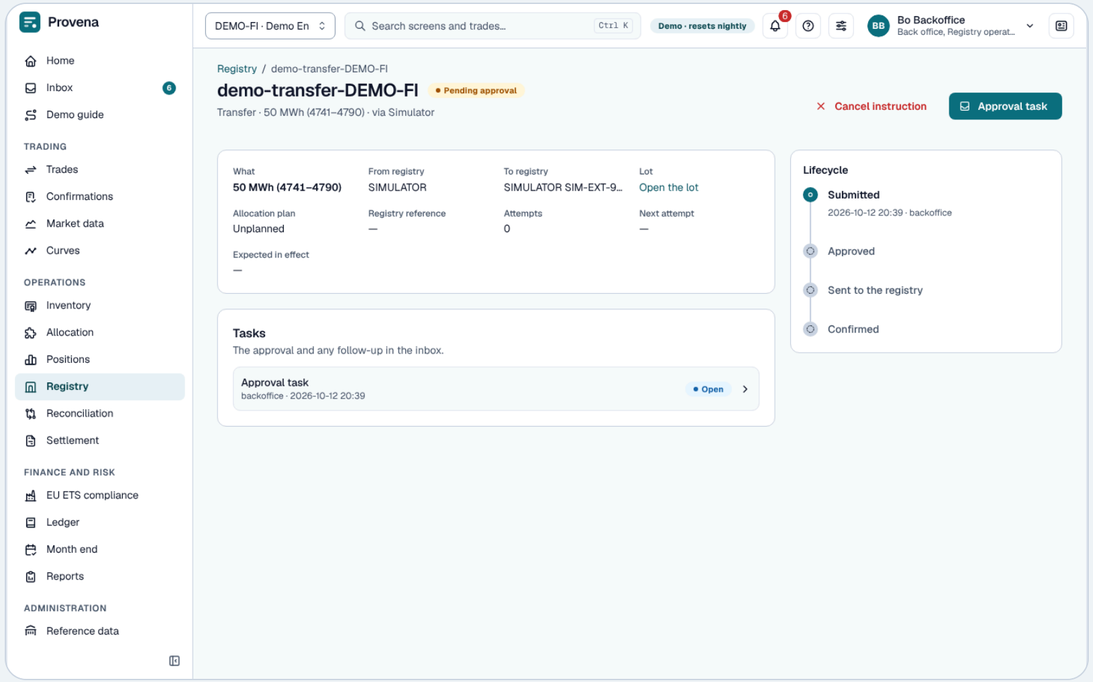
Specific certificates serve generic sales: allocated by value, with the give-away shown.

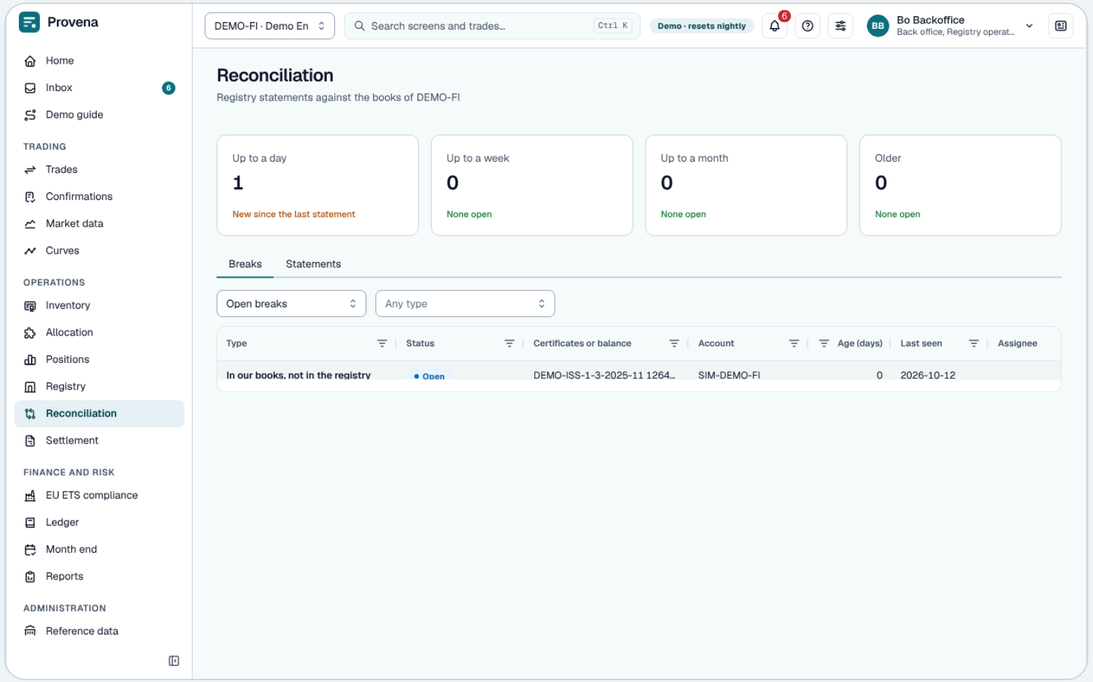
Available to sell, a product tree netted through eligibility, and future supply.

Portfolio value next to each run, and limits on concentration, expiry and basis.

REGISTRY AND RECONCILIATION

Every transfer accounted for





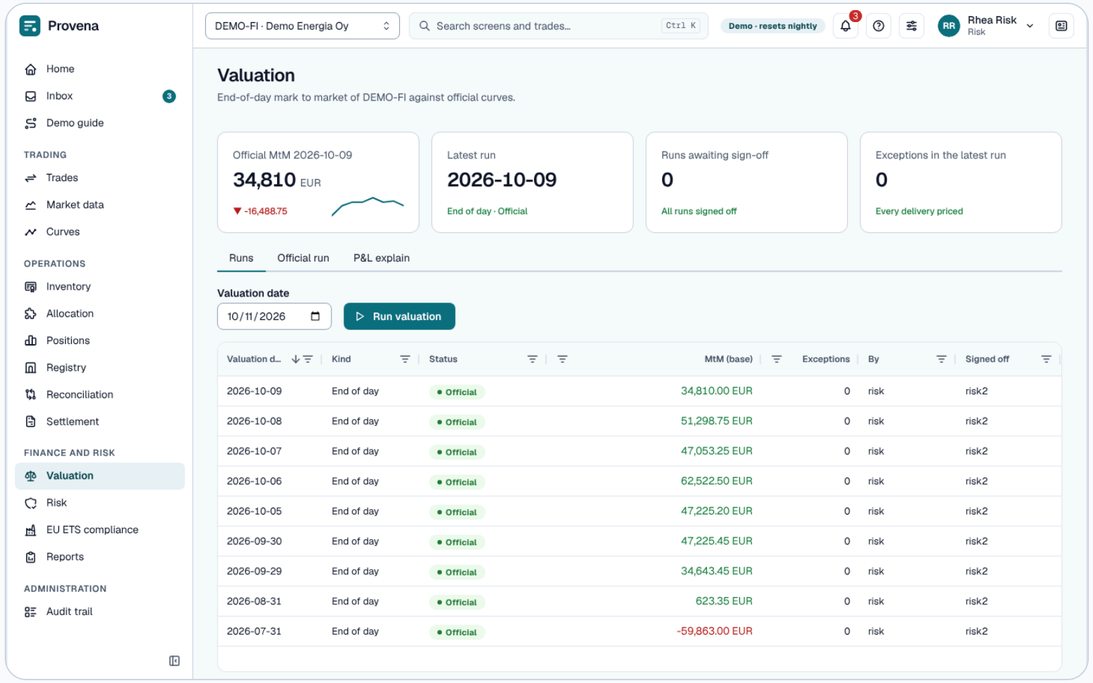
Instructions stored first, sent afterwards; retries are safe.

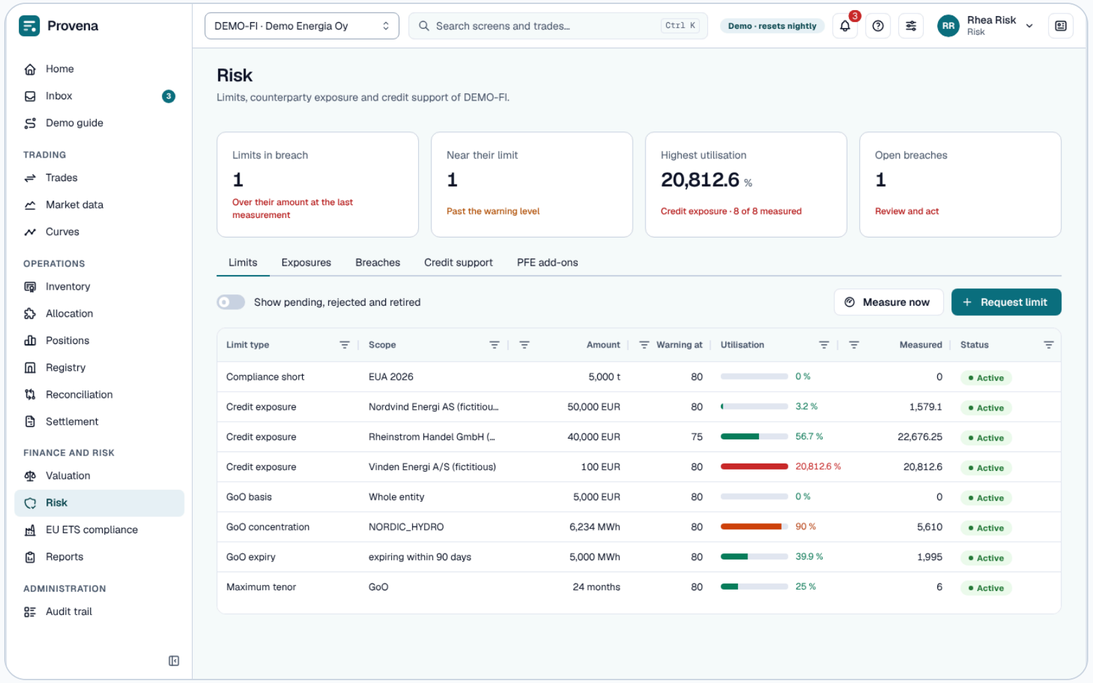
Registry statements reconciled with our books.

Each difference is a break with an owner and an outcome.

MARKET DATA, VALUATION AND RISK

Prices a second person signs off





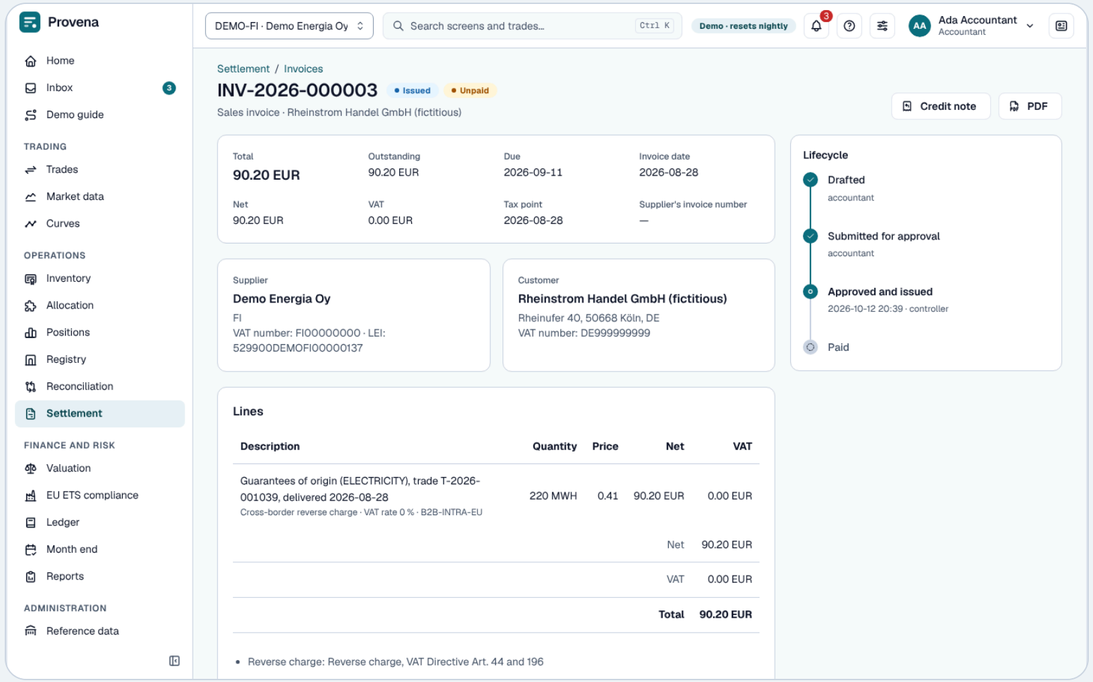
Curves become official once a second person signs them off.

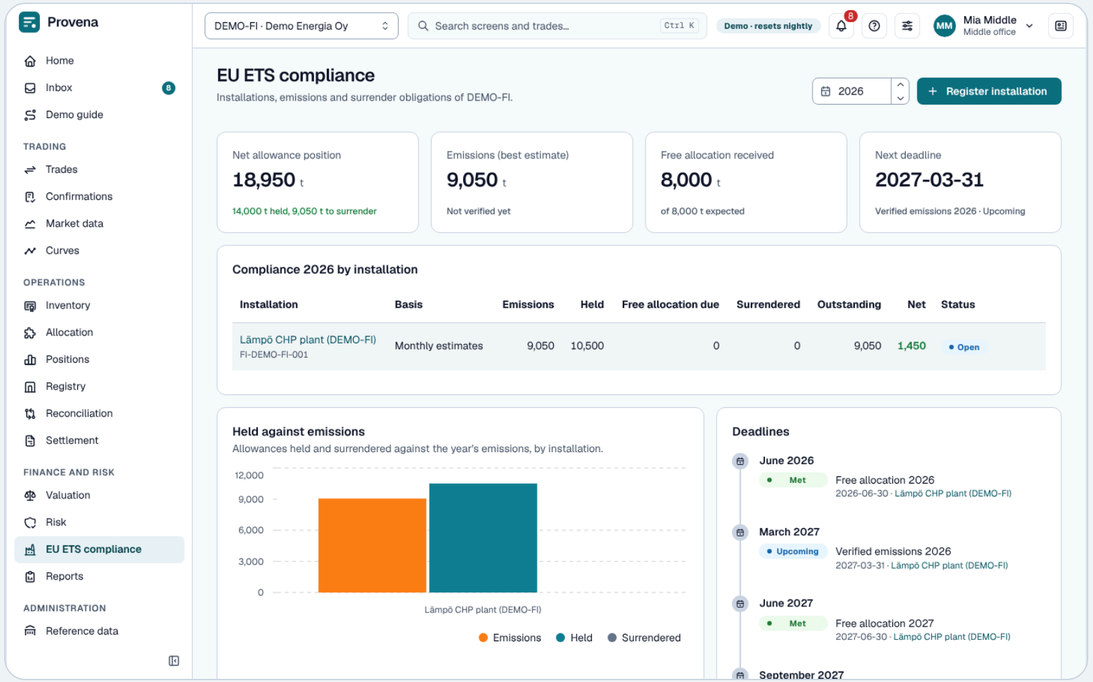
Official end-of-day valuation and P&L explain.

Limits, utilisation, breaches and credit support.

SETTLEMENT AND EU ETS

Invoices and compliance





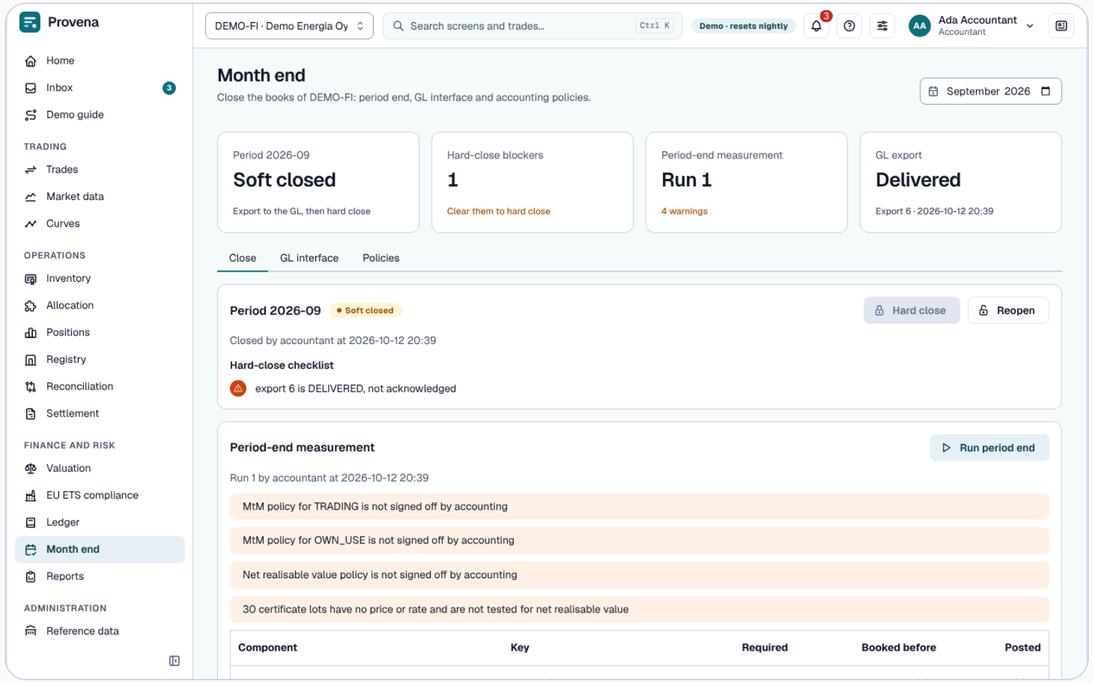
Deliveries become invoices, approved by a second accountant.

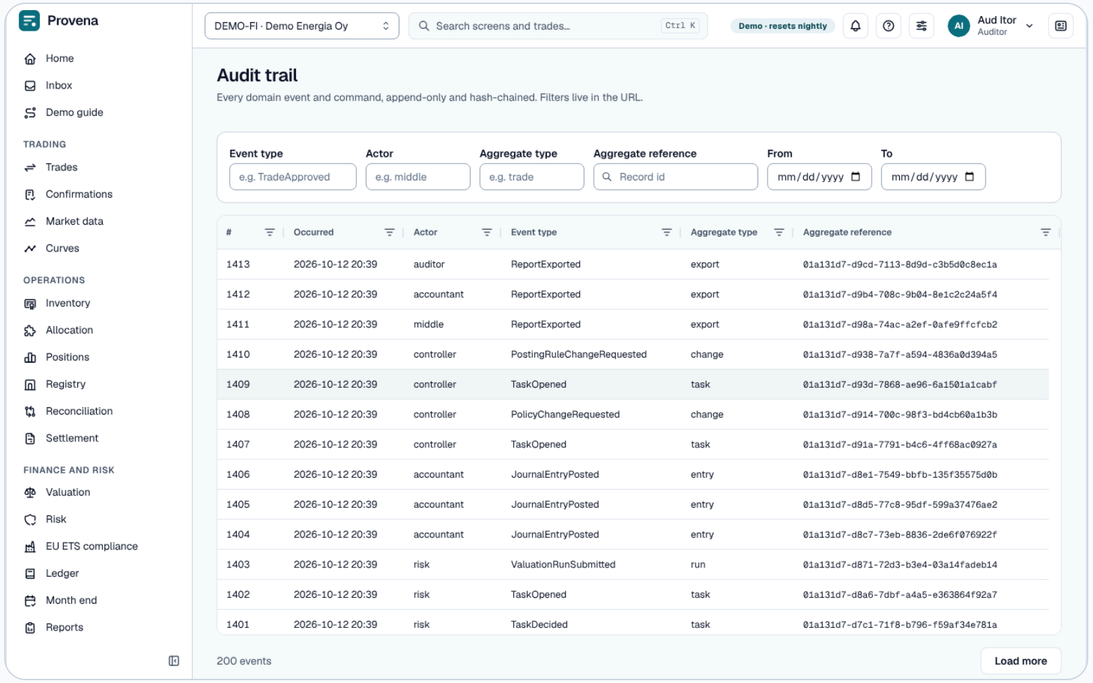
Payments, netting and VAT rules, reverse charge included.

Emissions, free allocation and surrenders per installation.

FINANCE AND CONTROL

Books an auditor can follow





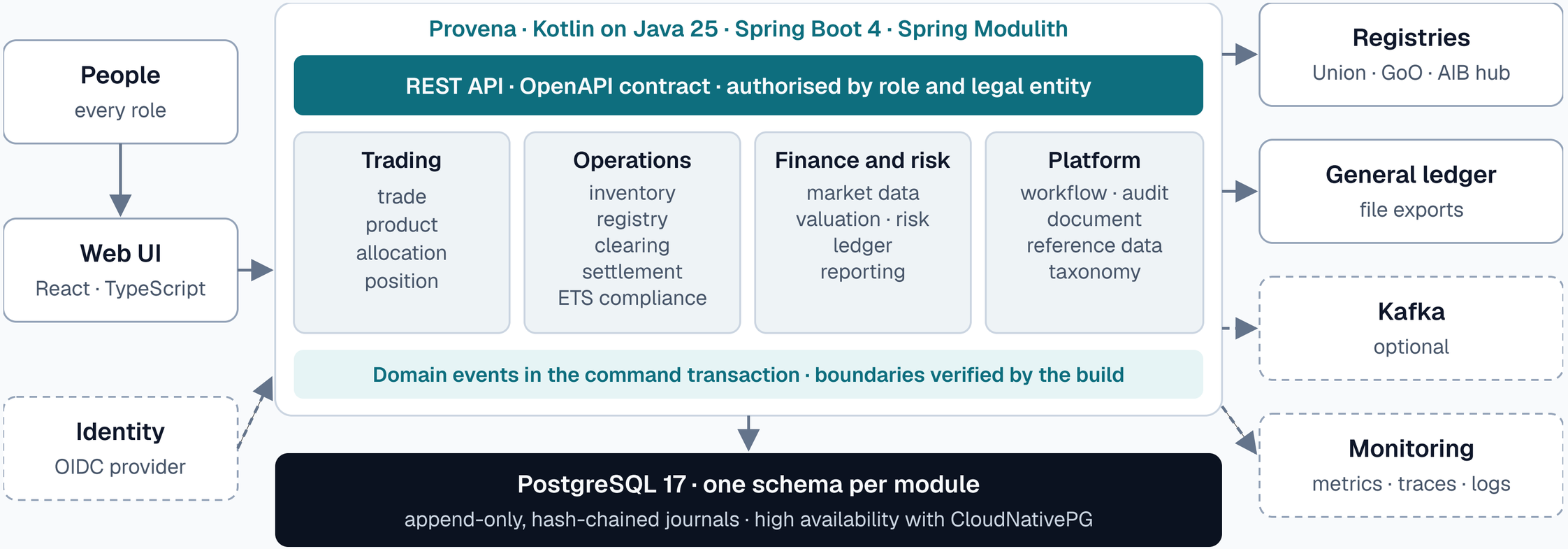
Posting rules feed an append-only, hash-chained journal.

Month end: period end, soft and hard close, GL export.

An audit trail of every change: who, when and why.

ARCHITECTURE

One application, nineteen bounded contexts



RELIABILITY

Designed to targets, measured, rehearsed

99.9 %
availability target in business hours

15 min
recovery point; recovery time 4 hours

556/s
requests under stress, 104 users, no errors

15 ms
read p95 at baseline load (target 300 ms)

Stateless, highly available

Scales on CPU or request rate; long runs are asynchronous; a registry outage pauses only that registry.

Recovery rehearsed

A point-in-time recovery drill on CloudNativePG restores a backup and verifies every hash chain.

Observable

Metrics, traces and structured logs; alert rules with a runbook each; a trace id on every response.

SECURITY

Least privilege, two people, a checked supply chain

Identity and access

Single sign-on with OIDC and audience checks. Every one of the 339 API operations is tested against a minimal-role user.

Segregation of duties

Four eyes on everything that moves value, in the services and again in the database, which the application reaches with a runtime role that cannot alter its tables.

Hardened web tier

Strict content security policy, CSRF protection, formula-safe exports; ZAP and accessibility checks.

Supply chain

SBOMs for every build; the image is scanned and a high or critical fixable flaw stops the release; dependencies locked and checksummed.

ROBUSTNESS AND DATA INTEGRITY

A system of record that cannot quietly drift

Append-only, hash-chained

Lot journal, ledger, allowance movements, audit trail and task history; verified every night. Errors are reversed, never edited.

Exactly-once effects

Registry calls through an outbox; idempotent deliveries and GL exports; a timeout is reconciled, not resent.

Reproducible as of any date

Bitemporal lots, prices and reference data; every report run is logged with its content hash.

Failures stay contained

A downstream problem is parked for attention; exact decimals only, no floating point in the domain.

MAINTAINABILITY

Boundaries and contracts the build enforces

Module boundaries

Each module owns its schema and exposes one API package; any other dependency fails the build.

Contracts

The OpenAPI document is committed; the web client's types are generated from it, never written by hand.

Decisions on record

55 architecture decision records, arc42 documentation, a contributor handbook and a user manual inside the application.

Strict by default

Warnings are errors; static analysis with no baselines, coverage floors that only rise, locked and verified dependencies; main moves only through green pull requests.

ROADMAP

Seventeen increments, each one shippable

Inc.	Delivered
1	Core domain: products and criteria, trades, inventory, allocation, positions, ledger, audit
2	Registry integration, reconciliation, settlement and invoicing, confirmations, approval inbox
3	Market data, curves, valuation, risk and credit, ETS compliance, period end, reporting, GL
4–5	Web UI for every area, including the operations screens
6	Operations at scale: observability, supply chain, Kubernetes, HA database, load tests, DR
7	Showcase: design system, six months of demo data, in-app manual and guided tours
8–10	GoO portfolio: value-based allocation, available to sell, future supply, limits; versioned releases; figures in under a second
11	Valuation depth: discount curves and discounted MtM, time and rates in the explain, cash flow hedge accounting
12	Futures margining: daily variation margin, clearing statements and breaks, margin calls, delivery at final settlement
13	Options on EUA futures: volatility curve, Black-76 with greeks, premiums, exercise at expiry, delta in risk
14–15	Production readiness after an independent review: access control, sealed chains, money seams, alerts, DR, gates
16–17	Review closed: asynchronous long runs, registry circuit, races and idempotency, repeatable journeys, no unchecked nulls

PATH TO PRODUCTION

What remains needs a partner, not more code

Assurance

- External penetration test and retest
- DR drill on a production backup
- ETS deadline rules checked by compliance
- UAT with real users; a load-test environment

Specifications

- Registry APIs (G-REX, NECS) and AIB code lists
- The receiving GL's file layout
- EMIR field layout and trade repository
- Market data vendors

Sign-offs

- Accounting policies: MtM by book type, NRV, provision
- VAT rules
- PFE add-ons

Next to build

The October 2026 review's code findings are all fixed (0.17.0). Candidates: minimum shares against available to sell, fair value hedges, a volatility smile, gas and hydrogen GoO, ETS2 allowances.

IN NUMBERS, AND TRY IT

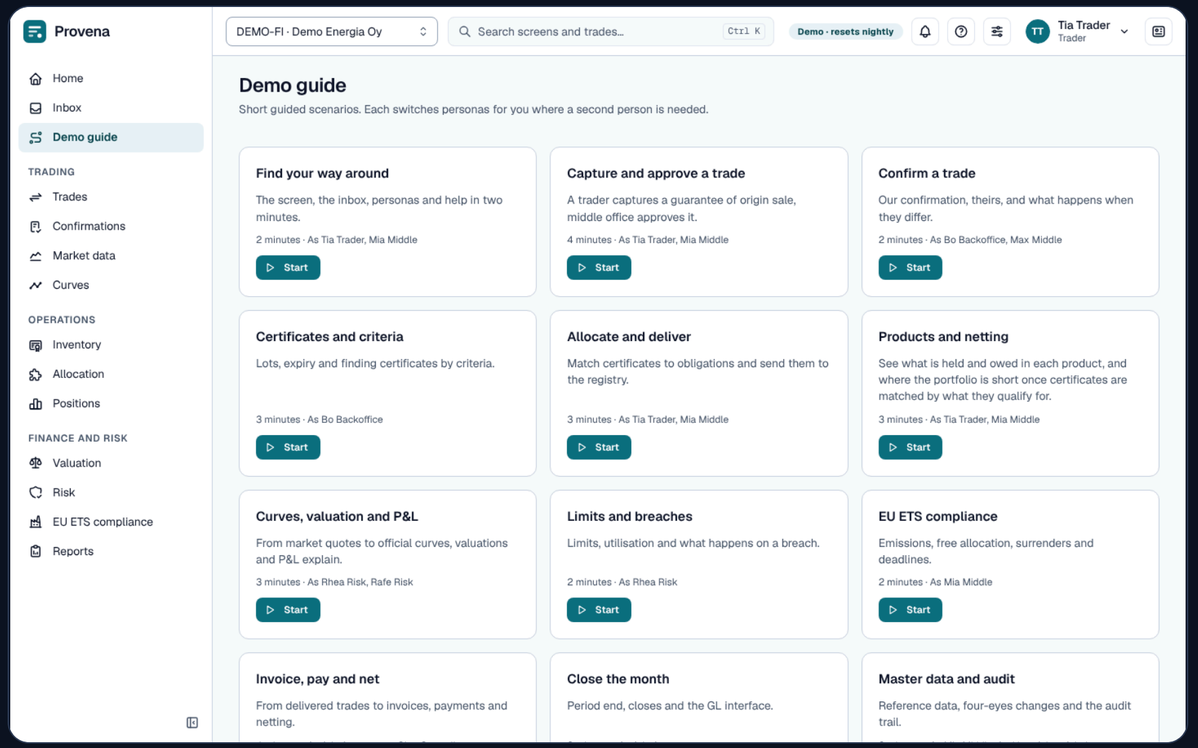
Evidence you can run yourself

489
backend unit and integration tests

454
web UI component tests

89
browser journeys with accessibility checks

313 / 339
API operations with a screen; the rest have a recorded reason



The demo with guided tours and a manual in the app, or your own copy with Docker in minutes.

Provena · 18 · Counted on the 0.17.0 build · github.com/Pianonurmi/goo-ets-etrm